REAL-TIME INDIAN SIGN LANGUAGE RECOGNITION USING EFFICIENT MULTI-FEATURE ATTENTION MECHANISM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Communication barriers between hearing-impaired individuals and the general population remain a significant social challenge. Indian Sign Language (ISL) is the primary communication medium for many deaf and hard-of-hearing individuals in India, but the lack of accessible translation tools often restricts interaction with non-signers. To address this problem, this work proposes a real-time Indian Sign Language recognition system based on an Efficient Multi-Feature Attention Mechanism.

The proposed system processes live video streams captured from a camera and converts sign gestures into meaningful text. Video frames are extracted and preprocessed to maintain uniform spatial and temporal representation. Spatial visual cues, such as hand shape, orientation, and appearance, are learned using an EfficientNet model pretrained on ImageNet through transfer learning. In parallel, temporal motion patterns are captured using a CNN–RNN architecture.

To improve recognition performance, an attention mechanism dynamically assigns importance to spatial and temporal features, emphasizing relevant gesture components while suppressing less informative information. The weighted features are fused and passed to a GRU-based sequence modeling network, and a fully connected softmax classifier predicts the corresponding ISL gesture class. Designed for real-time deployment, the framework maintains low computational overhead while delivering reliable recognition accuracy and provides a strong foundation for future work on continuous sign language translation.

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LIST OF ABBREVIATIONS

ISL	-	Indian Sign Language
CNN	-	Convolutional Neural Network
RNN	-	Recurrent Neural Network
BiLSTM	-	Bidirectional Long Short-Term Memory
GRU	-	Gated Recurrent Unit
CTC	-	Connectionist Temporal Classification
TTS	-	Text-to-Speech
RGB	-	Red Green Blue
TP	-	True Positive
TN	-	True Negative
FP	-	False Positive
FN	-	False Negative
F_1	-	F1 Score

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Communication is a fundamental requirement for social participation, education, healthcare access, and economic opportunity. For members of the deaf and hard-of-hearing community in India, Indian Sign Language (ISL) is a natural and expressive medium that encodes meaning through coordinated hand shape, hand orientation, movement trajectory, body posture, and facial expression. In practice, however, smooth communication between ISL users and non-signers is still limited by the availability of trained interpreters and by the absence of reliable real-time assistive tools in many public contexts.

The research community has shown that deep learning methods can significantly improve visual recognition tasks, including gesture and sign understanding. Modern computer vision models can learn robust spatial representations from raw RGB frames, while sequence models can capture temporal evolution across consecutive frames. These developments are strongly aligned with sign language understanding, where meaning depends not only on static hand configuration but also on motion continuity and contextual timing [4, 8, 9].

Despite this progress, ISL-focused systems still face practical and technical constraints. Gesture classes can be visually similar, signer-specific variations can be large, and unconstrained environments introduce noise from lighting, background clutter, and camera perspective. Therefore, there is a clear

need for an ISL recognition framework that is both accurate and computationally efficient, and that can operate with real-time responsiveness for deployment-oriented scenarios.

1.2 PROBLEM STATEMENT

Although existing sign language recognition approaches report encouraging results on benchmark datasets, many systems remain difficult to deploy in real-time communication settings. A common limitation is the imbalance between spatial and temporal modeling: models that focus heavily on frame-level appearance may miss motion continuity, whereas sequence-heavy models may lose fine-grained hand configuration cues. This trade-off directly affects recognition quality for signs that differ by subtle movement or short-duration transitions.

Another challenge is computational efficiency. High-capacity architectures may improve offline accuracy, but they often increase latency and memory usage, reducing practical usability for continuous camera input. For assistive communication, response delay must be low enough to preserve conversational flow. In addition, ISL-specific data diversity is still evolving, and variability across signer style, speed, and environmental conditions can reduce generalization when models are trained without robust feature selection and fusion strategies.

The core problem addressed in this thesis is therefore the design of a real-time ISL recognition system that can jointly model spatial detail and temporal dynamics while remaining computationally efficient. The system must maintain stable performance under realistic input variability and produce output that can directly support communication between ISL users and non-signers.

1.3 OBJECTIVES OF THE PROJECT

The proposed work is aimed at developing an assistive ISL recognition pipeline that is academically rigorous and practically deployable. In line with this goal, the project objectives are defined to balance algorithmic performance, model efficiency, and usability of generated output.

The specific objectives are as follows:

To design and implement a real-time ISL recognition architecture using RGB video input.

To extract discriminative spatial features using EfficientNet-based visual encoding.

To capture gesture dynamics through temporal modeling over frame sequences.

To integrate a multi-feature attention mechanism that emphasizes informative visual-temporal cues.

To generate usable communication output by converting recognized signs into readable text and speech.

Collectively, these objectives define a complete path from visual acquisition to assistive output, with emphasis on both recognition reliability and real-time behavior.

1.4 SCOPE OF THE WORK

This thesis focuses on the recognition of ISL gestures from RGB video streams using a deep learning pipeline that combines spatial feature extraction,

temporal modeling, and attention-based feature refinement. The implementation scope includes data preprocessing, model design, training and inference workflow, and output generation as text and speech. A key requirement within this scope is real-time oriented performance, so architectural choices are made with computational efficiency in mind rather than only maximizing offline benchmark accuracy.

The present work is intentionally bounded to maintain technical clarity and measurable outcomes. The system is developed primarily for isolated gesture recognition and structured evaluation under controlled experimental settings. While the framework can inform future extensions, full continuous sentence-level translation, multilingual sign integration, and edge deployment on low-power embedded hardware are treated as beyond the immediate scope of this thesis and are discussed as forward directions.

1.5 MOTIVATION

The primary motivation for this work is the need to reduce communication barriers faced by ISL users in everyday interactions. In academic institutions, hospitals, offices, and public service centers, communication often depends on interpreter availability, which may not always be practical. A robust automated recognition system can act as an assistive bridge, enabling faster and more inclusive interaction between signers and non-signers.

From a technical perspective, ISL recognition is a meaningful and challenging problem at the intersection of computer vision, sequence modeling, and human-centered AI. It requires the model to understand both fine-grained visual structure and temporal progression under real-world variability. Addressing this challenge contributes not only to assistive communication technology but also to broader research on efficient multimodal understanding systems.

This thesis is therefore motivated by a combined social and engineering objective: to build a system that is both practically useful and technically sound, with a clear pathway toward future real-world deployment.

1.6 OUTLINE OF THE THESIS

This thesis is organized into five chapters, each addressing a specific stage of the study. Chapter 1 introduces the communication context, problem definition, objectives, scope, and motivation for the proposed work. Chapter 2 reviews existing literature and related methods in sign language recognition, followed by a critical analysis of current limitations and research gaps. Chapter 3 presents the proposed methodology, including system architecture, module-level design, algorithmic flow, and supporting mathematical formulation.

Chapter 4 describes implementation details, dataset and input handling, experimental setup, and performance-oriented discussion of results. Finally, Chapter 5 summarizes the major contributions of the thesis, discusses practical limitations, and outlines directions for future enhancement. This progression is intended to provide a coherent transition from problem framing to methodological design, empirical validation, and concluding insights.

CHAPTER 2

RELATED WORK

2.1 OVERVIEW OF EXISTING SYSTEMS

Early sign language recognition systems relied on handcrafted features and traditional machine learning algorithms such as Hidden Markov Models (HMM) and Support Vector Machines (SVM). These approaches provided limited accuracy due to their inability to capture complex visual patterns.

With the emergence of deep learning, vision-based approaches using CNNs and RNNs have become dominant. These methods significantly improve recognition accuracy by learning hierarchical feature representations directly from data.

2.2 EXISTING METHODS AND TECHNIQUES

Modern approaches can be broadly categorized as follows:

Spatial feature-based methods: use CNN architectures such as EfficientNet to extract hand shape and appearance features [1].

Temporal sequence models: use RNN, LSTM, or GRU networks to model motion dynamics across frames.

Attention-based models: assign importance weights to relevant features and improve recognition performance.

Transformer-based models: capture long-range dependencies using self-attention [7].

Multimodal systems: combine RGB images with pose or depth data for improved accuracy.

2.3 ADVANTAGES AND LIMITATIONS OF EXISTING APPROACHES

Advantages:

High accuracy due to deep feature learning,

Ability to handle complex gesture patterns,

Improved temporal modeling.

Limitations:

High computational complexity,

Dependency on large datasets,

Limited focus on real-time performance,

Insufficient research on Indian Sign Language.

2.4 RESEARCH GAP IDENTIFIED

From the literature, the following gaps are identified:

Lack of efficient real-time ISL recognition systems,

Limited use of feature fusion techniques,

High computational overhead in advanced models,

Insufficient datasets and benchmarks for ISL.

These gaps highlight the need for a lightweight and efficient hybrid model for practical ISL recognition.

CHAPTER 3

PROPOSED METHODOLOGY AND SYSTEM DESIGN

3.1 OVERALL SYSTEM ARCHITECTURE

The overall architecture of the proposed Indian Sign Language recognition system is designed to process visual gesture input and convert it into meaningful textual and audio output. The system operates through a multi-stage deep learning pipeline that extracts both spatial and temporal information from the video sequence.

The complete workflow of the proposed system is illustrated in Fig. 3.1, which presents the architecture diagram of the model and the interaction between different processing modules. As shown in Fig. 3.1, the system begins with the acquisition of video input captured using a standard RGB camera. The input video is decomposed into individual frames through preprocessing. These frames are resized and normalized to maintain consistent spatial dimensions and improve numerical stability during model training and inference.

The processed frames are passed to the feature extraction stage where the system learns important visual characteristics of the gesture. Two complementary streams are used to capture different aspects of sign language gestures. The first stream performs spatial feature extraction using EfficientNet [1], which learns visual patterns such as hand shape, finger positioning, and gesture orientation. In parallel, the second stream performs temporal feature extraction

using a CNN–RNN framework that analyzes frame sequences to capture motion dynamics and gesture transitions over time.

After feature extraction, the system applies a multi-feature attention mechanism. This module assigns adaptive weights so the model focuses on informative frames and suppresses less relevant information. The weighted spatial and temporal features are combined using a feature fusion module to create a unified spatiotemporal representation.

The fused feature sequence is then processed by a sequence modeling component that captures contextual relationships between frames. A Connectionist Temporal Classification (CTC) layer aligns predicted labels with input sequences without requiring precise frame-level annotations. Finally, the recognized gesture sequence is converted into readable text and optionally transformed into speech output using a text-to-speech module.

Thus, the architecture integrates multiple deep learning components to effectively capture gesture appearance, motion dynamics, and contextual relationships, enabling accurate real-time recognition of Indian Sign Language gestures.

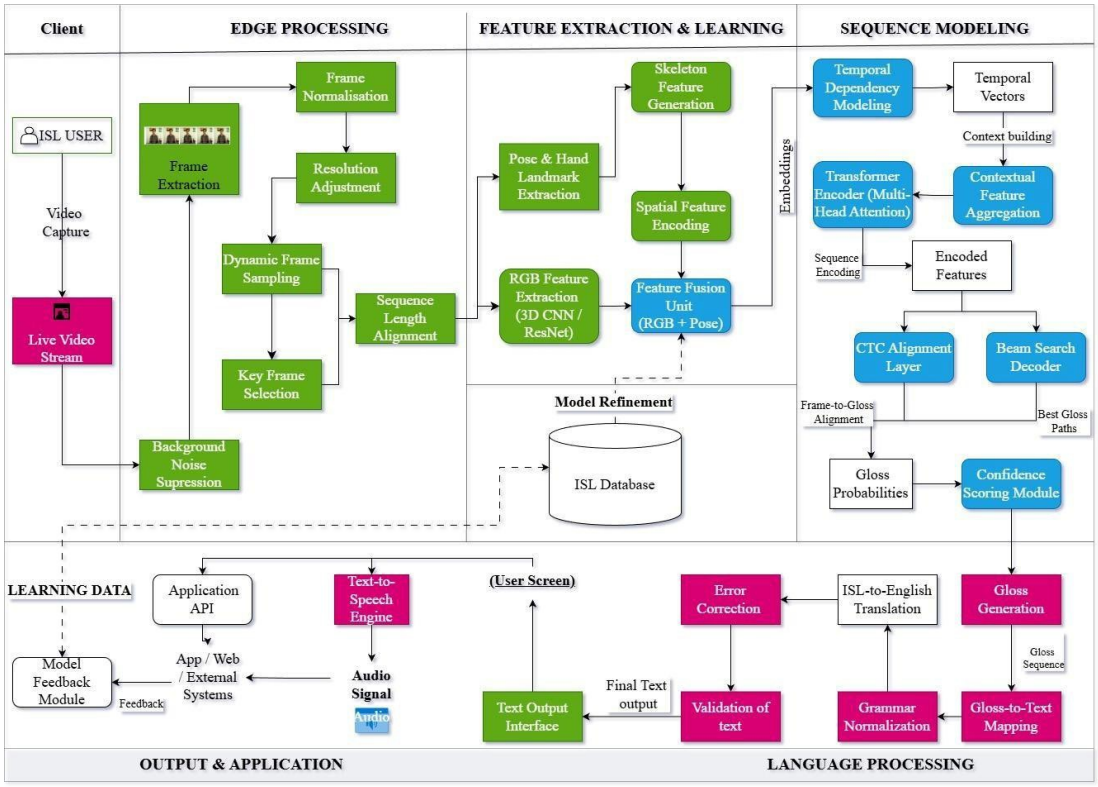

Figure 3.1: Architecture diagram of proposed model.

3.1.1 Block Diagram

3.1.1.1 Video Input Acquisition and Preprocessing

The system takes an RGB video of ISL gestures as input. Since raw video cannot be directly used, preprocessing is performed. The video is first converted into frames using uniform sampling to preserve temporal information. Each frame is resized and normalized to ensure consistent input size and stable training. Additionally, all sequences are standardized to a fixed length using padding or truncation, enabling efficient batch processing.

3.1.1.2 Spatial Feature Extraction Using EfficientNet

EfficientNet is used to extract spatial features from each frame, such as hand shape, orientation, and appearance. The model is pretrained and adapted through transfer learning, generating compact feature vectors with high representation quality and computational efficiency.

3.1.1.3 Temporal Feature Extraction Using CNN–RNN

To capture motion information, a CNN–RNN architecture is used. CNN extracts frame-level features, and RNN processes them sequentially to learn temporal dependencies, including gesture movement, direction, and transitions over time.

3.1.1.4 Multi-Feature Attention Mechanism

The attention mechanism assigns weights to spatial and temporal features based on their importance. It focuses on relevant frames and suppresses less useful information, improving recognition accuracy and robustness to noise.

3.1.1.5 Feature Fusion

Spatial and temporal features are combined to create a unified representation. This fusion captures both visual appearance and motion dynamics of gestures, making the model more effective in recognizing complex signs.

3.1.1.6 Sequence Modeling Using Transformer

The fused features are passed to a Transformer encoder [7], which models long-range dependencies using self-attention and captures relationships between frames efficiently.

3.1.1.7 Temporal Alignment Using CTC

CTC is used to align predicted outputs with input sequences without requiring exact frame-level labeling. This makes the system flexible and suitable for sequence-based gesture recognition.

3.1.1.8 Decoding Using Beam Search

Beam search is applied during prediction to select the most probable output sequence by evaluating multiple candidate sequences, improving overall accuracy compared to simple greedy decoding.

3.1.1.9 Language Processing and Grammar Refinement

The predicted gesture sequence is converted into text using mapping techniques. Grammar refinement is applied to correct sentence structure and improve readability.

3.1.1.10 Output Generation and Text-to-Speech Conversion

The final output is displayed as text and also converted into speech using a text-to-speech system. This enables effective communication between sign language users and non-signers.

3.2 MODULES DESCRIPTION

3.2.1 Video Acquisition and Preprocessing

3.2.1.1 Frame Extraction and Normalization

This module is responsible for converting raw video input into a structured and standardized format suitable for deep learning models. The system accepts an RGB video stream captured using a standard camera. Since deep learning models cannot directly operate on continuous video streams, the video is first decomposed into individual frames.

Frame extraction is performed using uniform sampling, where a fixed number of frames are selected from each video sequence. Uniform sampling preserves temporal information while avoiding redundant frames that do not contribute meaningful information. This step is especially important for real-time processing because it reduces computational overhead without losing critical motion cues.

Once extracted, each frame is resized to a predefined spatial resolution compatible with convolutional neural networks. Frame normalization is then applied by scaling pixel intensity values to a standard range. Normalization improves numerical stability during training and ensures consistent feature learning across videos captured under varying lighting conditions. Finally, all frame sequences are standardized to a fixed temporal length using padding or truncation, enabling efficient batch processing during training and inference.

Output: A fixed-length sequence of normalized RGB frames.

3.2.1.2 Pose and Landmark Extraction

This sub-module extracts structured motion and positional information from video frames. From each frame, key body, hand, and facial landmarks are detected to represent the signer's movements in a compact and meaningful form. These landmarks encode geometric relationships such as joint positions, hand orientation, and finger movement patterns.

Pose and landmark information provides complementary motion cues that are invariant to background noise and illumination changes. By representing gestures in terms of relative positions and movements, the system improves robustness to environmental variations. The extracted landmark features are temporally aligned with the corresponding RGB frames to preserve synchronization between visual appearance and motion information.

Output: A sequence of landmark feature vectors aligned with video frames.

The end-to-end preprocessing pipeline is shown in Fig. 3.2.

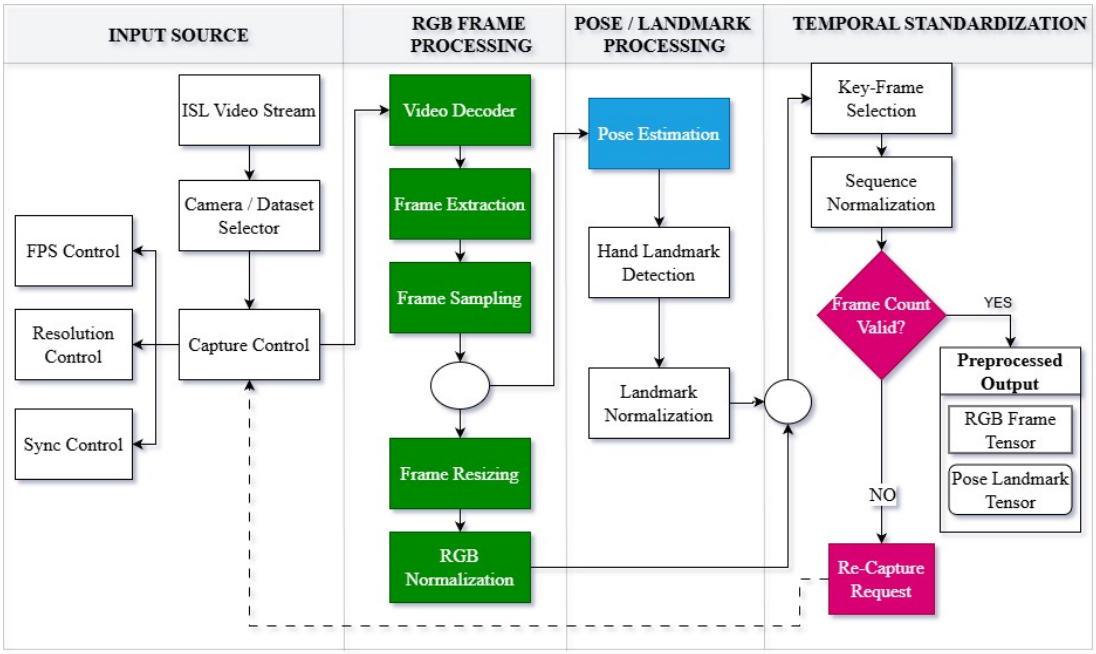

Figure 3.2: Video Acquisition and Preprocessing Module.

3.2.2 Feature Extraction and Fusion

3.2.2.1 RGB Feature Extraction Using CNN

This module extracts spatial appearance features from RGB frames using a convolutional neural network (CNN). Each normalized frame is passed through multiple convolutional layers that learn hierarchical visual representations. Early layers capture low-level patterns such as edges and textures, while deeper layers learn high-level semantic features such as hand shape, orientation, and configuration.

The CNN transforms each frame into a compact feature vector that encodes visual characteristics relevant to sign language gestures. These features capture fine-grained appearance details that are critical for distinguishing visually similar signs.

Output: Frame-wise RGB feature vectors.

3.2.2.2 Pose Feature Extraction and Attention-Based Fusion

This sub-module processes the landmark feature vectors extracted in the preprocessing stage and integrates them with RGB features using an attention-based fusion mechanism. Pose features capture motion trajectories and relative positional changes, while RGB features provide detailed appearance information.

An attention mechanism is applied to dynamically assign importance weights to both RGB and pose features. During training, the model learns which feature type contributes more to recognizing a particular sign. For example, signs with subtle finger movements may rely more on landmark features, while signs with distinctive hand shapes may rely more on RGB features.

The attention-based fusion module suppresses irrelevant or noisy features and emphasizes discriminative information. The weighted RGB and pose features are then combined to form a unified spatiotemporal representation.

Output: Fused feature sequence containing both appearance and motion information.

The feature extraction and fusion stage is presented in Fig. 3.3.

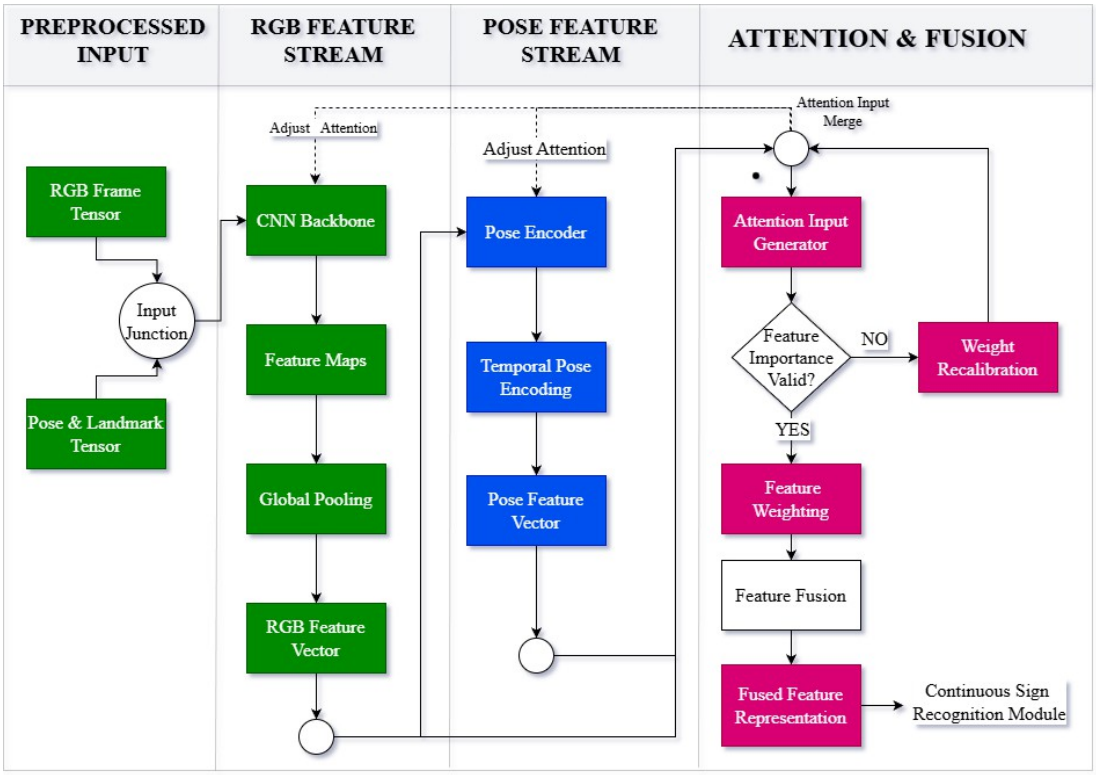

Figure 3.3: Feature Extraction and Fusion Module.

3.2.3 Continuous Sign Recognition

3.2.3.1 Temporal Modeling Using Transformer or BiLSTM

This module models the temporal evolution of the fused feature sequence. Since sign language gestures unfold over time, temporal modeling is essential for understanding gesture progression and context.

Two alternatives are considered:

BiLSTM: processes the feature sequence in forward and backward directions to capture past and future dependencies.

Transformer: uses multi-head self-attention to model long-range dependencies across the sequence.

Temporal modeling enables the system to learn motion patterns, transitions between gesture components, and contextual relationships between frames. This is particularly important for continuous sign recognition, where multiple signs appear in a single video sequence.

Output: Context-aware temporal feature sequence.

3.2.3.2 Sign Prediction Using CTC Decoder

The temporally modeled feature sequence is passed to a Connectionist Temporal Classification (CTC) decoder. CTC enables sequence-to-sequence learning without requiring explicit alignment between frames and output labels.

During training, the CTC loss function allows the model to learn the most probable label sequence corresponding to the input feature sequence. During inference, a decoding strategy such as beam search is applied to generate the final predicted sign or sign sequence.

CTC is particularly effective for continuous sign recognition because it handles variable-length input and output sequences and naturally accounts for timing variations in gesture execution.

Output: Predicted sign or sign sequence.

The recognition module behavior is illustrated in Fig. 3.4.

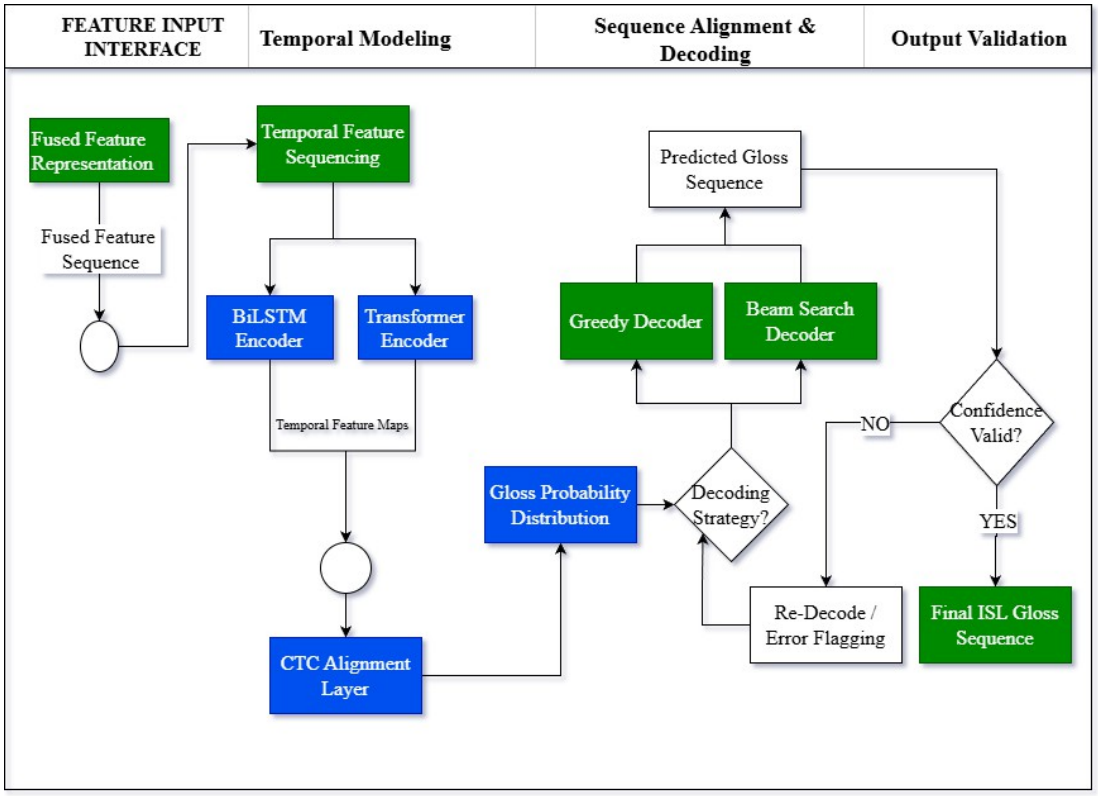

Figure 3.4: Continuous Sign Recognition Module.

3.2.4 Language and Audio Output

3.2.4.1 Sentence Correction and Caption Buffer

The predicted sign sequence is converted into textual form using a gloss-to-text mapping mechanism. Since the grammatical structure of sign language differs from spoken language, the generated text may contain syntactic inconsistencies.

A sentence correction module refines the generated text by correcting word order, grammatical errors, and incomplete phrases. A caption buffer is maintained to accumulate recognized signs over time, enabling smooth sentence formation in continuous recognition scenarios.

Output: Grammatically corrected text sentences.

The language and audio pipeline is shown in Fig. 3.5.

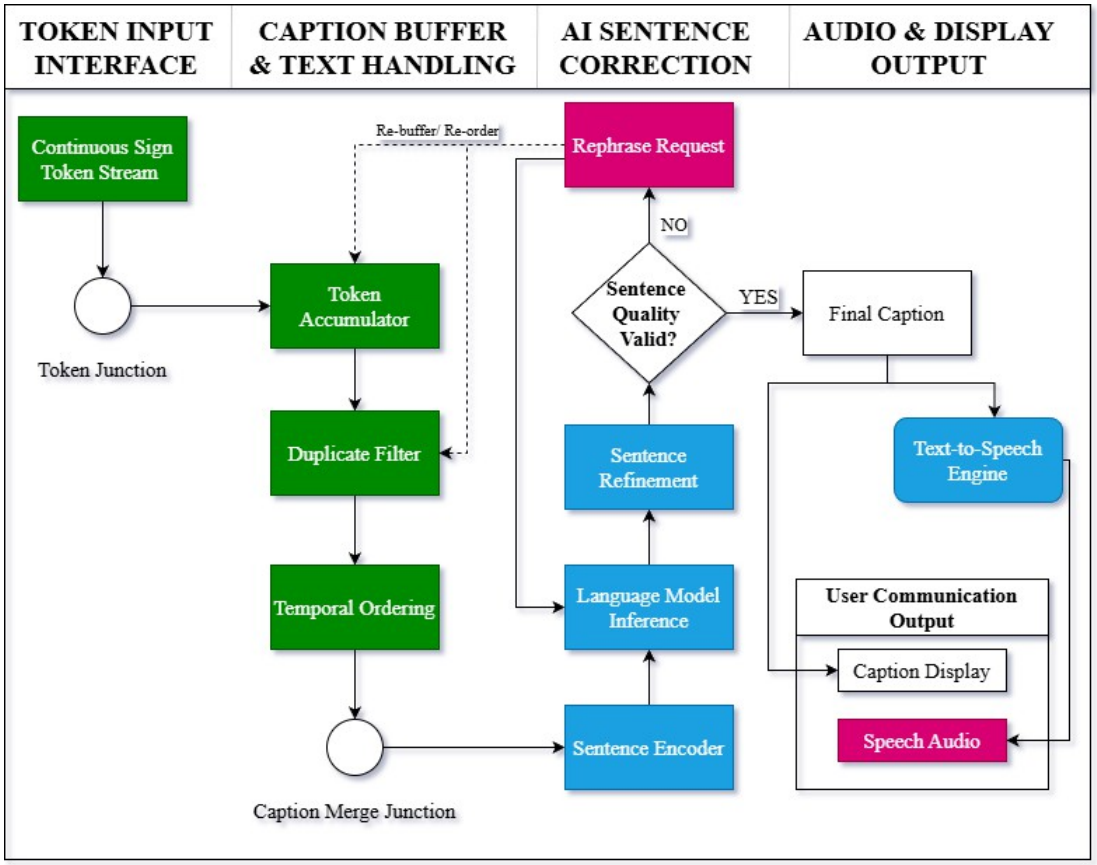

Figure 3.5: Language and Audio Output Module.

3.2.4.2 Text-to-Speech Output

In the final module, corrected text is converted into audible speech using a text-to-speech engine. The TTS system generates natural-sounding audio corresponding to the recognized sign language content. This module enables real-time verbal communication between signing users and non-signing individuals.

Output: Spoken audio output.

3.3 ALGORITHM

3.3.0.1 Step 1: Capture Input Video Stream

Capture a real-time RGB video stream from camera input representing sign gestures.

3.3.0.2 Step 2: Frame Extraction and Preprocessing

Extract frames using uniform sampling, resize and normalize each frame, and standardize sequence length with padding or truncation.

3.3.0.3 Step 3: Spatial Feature Extraction using CNN

Pass each frame through EfficientNet-based CNN layers to extract spatial features.

3.3.0.4 Step 4: Temporal Feature Extraction using RNN

Feed frame-level features to an RNN model (GRU/LSTM) to capture temporal dependencies.

3.3.0.5 Step 5: Apply Attention Mechanism

Apply attention to emphasize informative spatial-temporal features and suppress noisy data.

3.3.0.6 Step 6: Feature Fusion

Combine weighted spatial and temporal features to form a unified spatiotemporal representation.

3.3.0.7 Step 7: Sequence Modeling

Use a Transformer sequence model to learn long-range contextual relationships across frames.

3.3.0.8 Step 8: Gesture Prediction

Classify processed features into gesture classes using a fully connected layer with softmax activation.

3.3.0.9 Step 9: Output Generation (Text and Speech)

Convert predicted gestures into readable text and optionally generate speech using text-to-speech.

3.4 MATHEMATICAL MODEL / EQUATIONS

There are many criteria that can be used to evaluate the performance of a sign language recognition model. For isolated sign recognition, the commonly used metric is accuracy. Precision, recall, and F1-score are also widely used as complementary metrics. The performance metrics used in this study are given in Equations (1)–(4).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.3}$$

$$F_1 \text{ Score} = \frac{2TP}{2TP + FP + FN} \tag{3.4}$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

3.5 WORKFLOW / PROCESS FLOW

The system workflow is:

Video Input → Preprocessing → Feature Extraction → Attention → Feature Fusion → Sequence Modeling → Prediction → Output

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 IMPLEMENTATION ENVIRONMENT

The proposed Indian Sign Language recognition system is implemented using modern deep learning frameworks and tools. Development is carried out in a Python environment due to broad support for machine learning libraries.

Programming language: Python

Libraries used: TensorFlow or PyTorch, OpenCV, NumPy, and Pandas

Deep learning models: EfficientNet, CNN–RNN, Transformer

Development tools: Jupyter Notebook or Google Colab

Hardware requirements: GPU-enabled system for faster training and inference

These tools enable efficient implementation of video processing, feature extraction, and model training.

4.2 DATASET / INPUT DESCRIPTION

The system uses a video-based dataset consisting of Indian Sign Language gestures. Each sample in the dataset represents a gesture performed by a signer.

Input data consists of RGB video sequences,

Videos are converted into frames for processing,

Each sequence represents a single gesture class (isolated gestures),

Preprocessing ensures uniform frame size and sequence length.

The dataset and input representation are shown in Fig. 4.1.

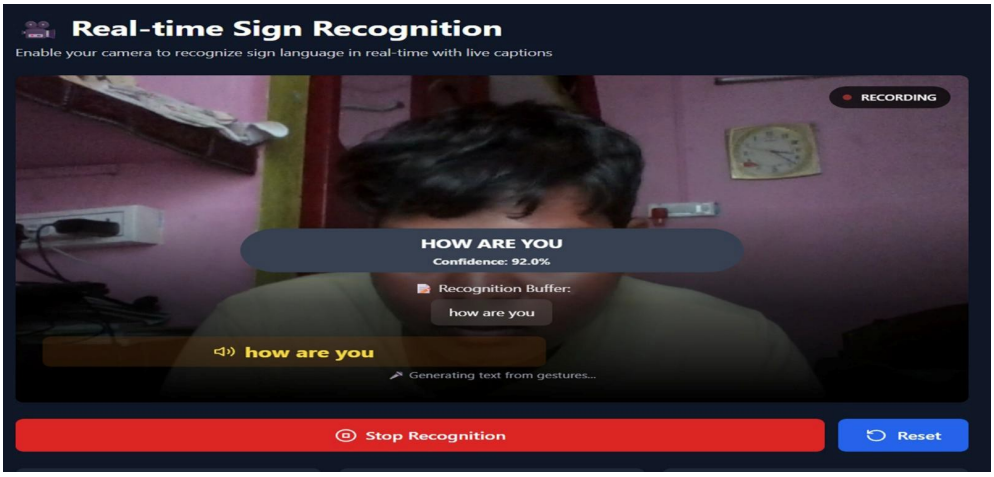

Figure 4.1: Dataset and input representation snapshot.

4.3 TEST CASES

The system is evaluated using multiple test cases to verify its performance under different input conditions. Test cases ensure that the system performs reliably across different scenarios. The test setup and evaluation example are shown in Fig. 4.2.

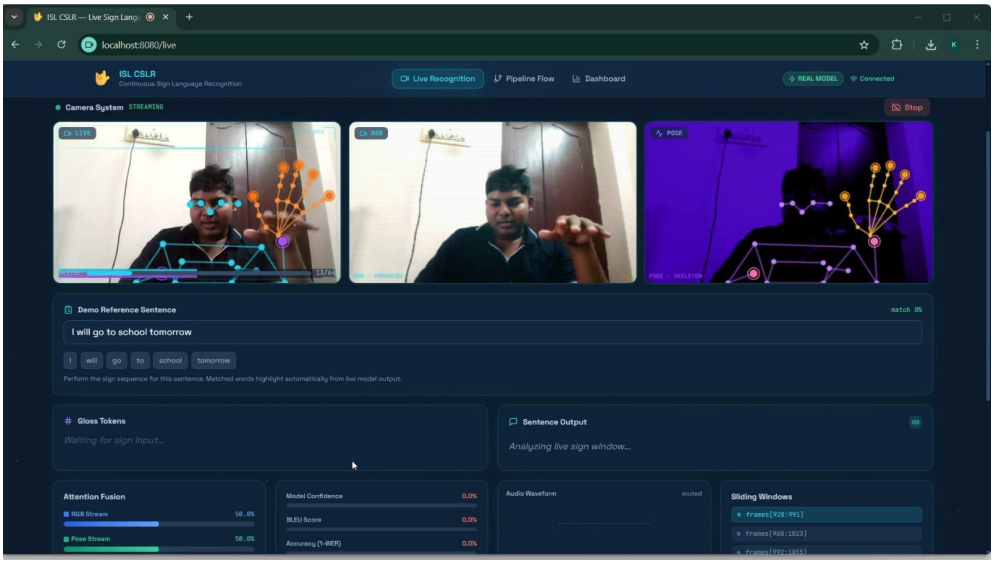

Figure 4.2: Test case setup and evaluation snapshot.

4.4 RESULTS AND DISCUSSION

The proposed system successfully recognizes Indian Sign Language gestures using a combination of spatial and temporal features.

4.4.1 Key Observations

EfficientNet effectively captures spatial features such as hand shape.

CNN–RNN captures motion dynamics across frames.

Attention mechanism improves feature selection and accuracy.

Transformer enhances sequence understanding.

4.4.2 Performance Metrics

The system is evaluated using:

Accuracy,

Precision,

Recall,

F1-score.

The model achieves good recognition accuracy for isolated gestures while maintaining computational efficiency.

4.4.3 Intermediate Outputs

The system produces several intermediate outputs:

Extracted frames from video,

Feature representations,

Predicted gesture labels,

Text output,

Speech output.

Representative intermediate outputs are shown in Fig. 4.3 and Fig. 4.4.

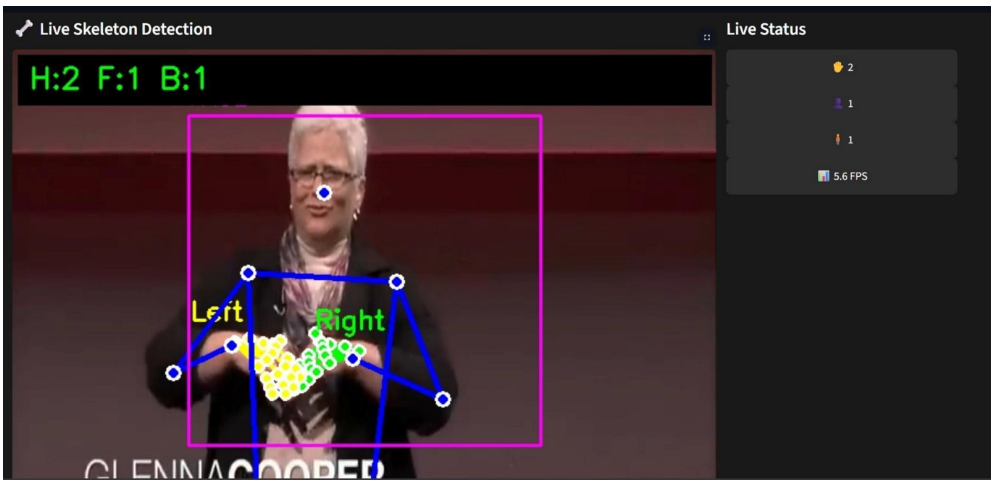

Figure 4.3: Intermediate output snapshot 1.

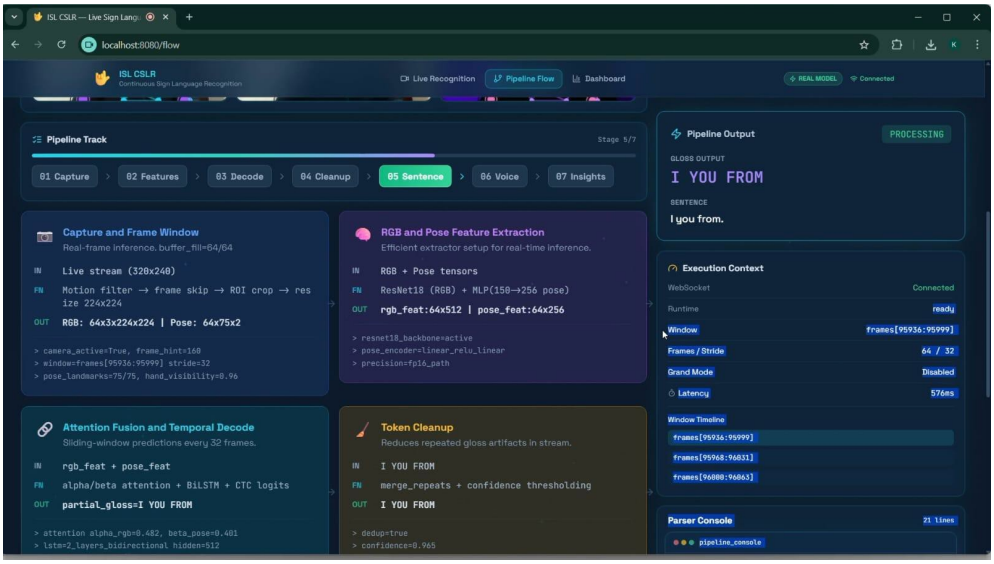

Figure 4.4: Intermediate output snapshot 2.

4.4.4 Result Analysis

The system performs well for clear and well-defined gestures.

Slight variations may occur in complex or fast gestures.

Performance is robust under moderate background noise.

The final result analysis snapshot is shown in Fig. 4.5.

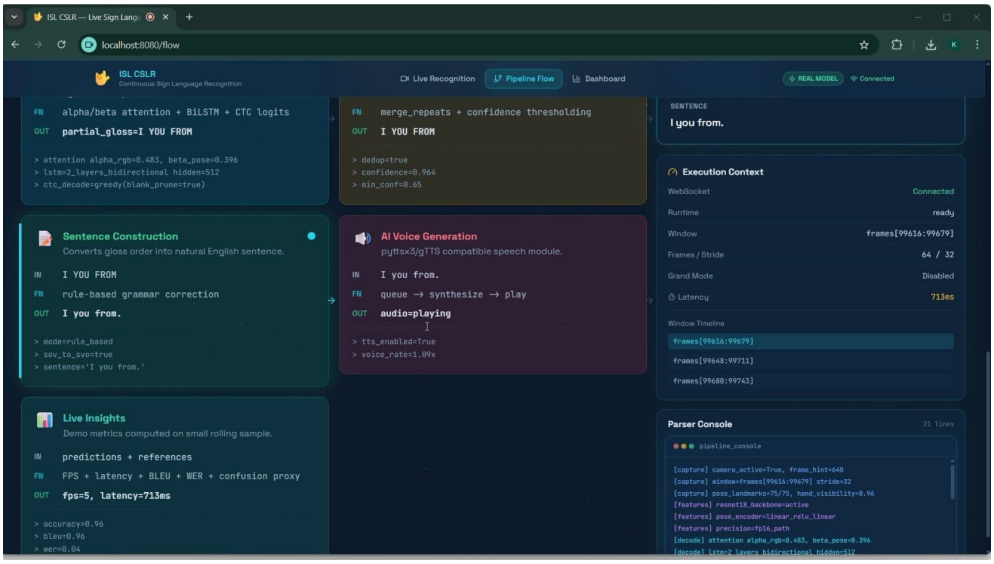

Figure 4.5: Result analysis snapshot.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 SUMMARY OF THE WORK

This project presented the design and implementation of a real-time Indian Sign Language (ISL) recognition system using deep learning techniques. The proposed system processes RGB video input and extracts both spatial and temporal features to accurately recognize hand gestures.

Spatial features are extracted using EfficientNet, while temporal dependencies are captured using a CNN–RNN architecture. A multi-feature attention mechanism is incorporated to improve feature selection by focusing on relevant gesture components. The system further uses a Transformer model for sequence modeling and a CTC layer for temporal alignment. The final output is generated in both text and speech form, enabling effective communication between sign language users and non-signers.

5.2 ACHIEVEMENTS

The major achievements of this project include:

Successful development of a real-time ISL recognition system,

Integration of spatial and temporal feature extraction techniques,

Implementation of an attention-based feature fusion mechanism,

Use of advanced models such as EfficientNet and Transformer,

Generation of text and speech output for user interaction,

Good accuracy with computational efficiency for isolated gestures.

5.3 LIMITATIONS

Despite promising results, the system has certain limitations:

The system is limited to isolated gesture recognition and does not support continuous sentence-level translation,

Performance depends on dataset quality and size,

Computational requirements can be high for low-end devices,

Accuracy may reduce for complex gestures or noisy environments.

5.4 FUTURE WORK

The proposed system can be further enhanced in the following ways:

Extension to continuous sign language recognition and full sentence translation,

Integration of multimodal inputs such as pose landmarks and depth data,

Development of a mobile or embedded application for real-world deployment,

Improvement of dataset size and diversity for better generalization,

Incorporation of advanced Transformer-based architectures for higher accuracy.

APPENDIX 1

ADDITIONAL MATERIAL

1.1 REPRODUCIBILITY CHECKLIST

The following checklist summarizes reproducibility controls used in this project:

fixed data preprocessing sequence (decode, sample, resize, normalize),

explicit module boundaries between extraction, fusion, decoding, and output,

consistent metric definitions for recognition, sequence, and translation quality,

project-level scripts for environment setup and service startup,

integrated figure and table assets stored directly inside the thesis repository.

1.2 CONFIGURATION SNAPSHOT

Table A 1.1: Representative configuration snapshot

Configuration group	Representative setting
Input pipeline	RGB video stream with frame downsampling profile from report artifacts
Feature extraction	Dual-stream processing: appearance cues + pose/motion cues
Temporal modeling	Sequence encoder with CTC-aligned decoding
Output stage	Token buffering, caption refinement, optional TTS synthesis
Evaluation output	Accuracy/F1, WER/CER, BLEU/ROUGE, deployment indicators

1.3 ADDITIONAL NOTES ON DEPLOYMENT

Report evidence indicates that deployment quality depends strongly on three practical controls: capture quality, vocabulary coverage, and signer diversity in training. For production-scale deployment, periodic data refresh and calibration checks are recommended.

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